

McBRAINSEY & CO.

INDEPENDENT REPORT

TUNEWORCS ENTERTAINMENT AGENCY



GROUP 2

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Phase 1: Data Familiarization & Documentation

1. Read all tables

AGENTS	People who work for the company and manage entertainer bookings.
CUSTOMERS	Individuals or organizations who hire entertainers.
MEMBERS	Individual members of entertainment groups or bands.
ENTERTAINERS	Performers or entertainment groups.
ENGAGEMENTS	A booking or scheduled performance event.
MUSICAL_STYLES	Types or genres of music (e.g., jazz, rock, classical).
ENTERTAINER_STYLES	Links an entertainer to the musical styles they perform.
MUSICAL_PREFERENCES	Customer preferences for musical styles.
ENTERTAINER_MEMBERS	Relationship between entertainers and their individual members.
ZTBLDAYS	Calendar dates.
ZTBLMONTHS	Months and related date calculations.
ZTBLSKIPLABELS	Control or configuration value for label skipping.
ZTBLWEEKS	Calendar weeks.

2. Primary keys & Foreign keys

Table	Primary Key(s)	Foreign Key(s)
Agents	AgentID	—
Customers	CustomerID	—
Members	MemberID	—
Entertainers	EntertainerID	—

Engagement	EngagementNumber	CustomerID → CustomersAgentID → AgentsEntertainerID → Entertainers
Musical_Styles	StyleID	—
Entertainer_Styles	EntertainerID, StyleID	EntertainerID → EntertainersStyleID → Musical_Styles
Musical_Preferences	CustomerID, StyleID	CustomerID → CustomersStyleID → Musical_Styles
Entertainer_Members	EntertainerID, MemberID	EntertainerID → EntertainersMemberID → Members
ztblDays	DateField	—
ztblMonths	YearNumber, MonthNumber	—
ztblSkipLabels	LabelCount	—
ztblWeeks	WeekStart	—

3. Data Dictionary

Table: Agents

Agents Table		
AgentID	Unique identifier for each booking agent	
AgtFirstName	Agent's first name	varchar
AgtLastName	Agent's last name	varchar
AgtStreetAddress	Street address of the agent	varchar
AgtCity	City component of the agent's address	varchar
AgtState	State or region code	varchar
AgtZipCode	Postal/ZIP code	varchar
AgtPhoneNumber	Agent's contact phone number	varchar

DateHired	When the agent was hired	date
Salary	The agent's base salary	decimal
CommissionRate	Percentage or rate of commission	float

Table: Customers

Customers Table		
CustomerID	Unique ID for each customer (serial)	
CustFirstName	Customer's first name	varchar
CustLastName	Customer's last name	varchar
CustStreetAddress	Customer's mailing address	varchar
CustCity	Customer's city	varchar
CustState	State/region code	varchar
CustZipCode	Postal code	varchar
CustPhoneNumber	Customer's primary phone number	varchar

Table: Members

Members Table		
MemberID	Unique identifier for each band member	serial
MbrFirstName	Member's first name	varchar
MbrLastName	Member's last name	varchar
MbrPhoneNumber	Member's contact number	varchar
Gender	Member's gender code	varchar

Table: Entertainers

Entertainer Table		
EntertainerID	Unique identifier for each performer or group	serial
EntStageName	Public or stage name	varchar
EntSSN	Tax or government ID	varchar
EntStreetAddress	Mailing address	varchar
EntCity	City	varchar
EntState	State code	varchar
EntZipCode	ZIP/postal code	varchar
EntPhoneNumber	Entertainer's contact phone	varchar
EntWebPage	Website or profile page	varchar
EntEmailAddress	Entertainer's email	varchar
DateEntered	Date added to system	date

Table: Engagements

Engagement Table		
EngagementNumber	Unique ID for each entertainment booking	serial
StartDate	Start date of engagement	date
EndDate	End date of engagement	date
StartTime	Start time of engagement	time
StopTime	End time of engagement	time
ContractPrice	Total fee for the engagement	decimal
CustomerID	Linked customer	int
AgentID	Linked agent	int
EntertainerID	Linked entertainer	int

Table: Musical_Styles

Musical Styles Table		
StyleID	Unique music style ID	Serial
StyleName	Name of style or genre	varchar

Table: Entertainer_Styles

Entertainer Styles Table		
EntertainerID	Identifier for each performer or entertainment group	Int
StyleID	Music style ID	int
StyleStrength	Data type: smallint; Business meaning: Strength or rating Valid values: small integer (e.g., 1–10)	

Table: Musical_Preferences

Musical Preferences Table		
CustomerID	ID of the Customer who has this preference	int
StyleID	ID of the Style preferred by the customer	int
PreferenceSeq	Rank or sequence (1 = top preference)	smallint

Table: Entertainer_Members

Entertainer_Members Table		
EntertainerID	Identifier for each performer or entertainment group	int
MemberID	Identifier for each band member or group	int
Status	Membership status as 1 or 0	smallint

Table: ztblDays

ztblDays Table		
DateField	Calendar date	date

Table: ztblMonths

ztblMonths Table		
MonthYear	Text description of month/year; Valid values: e.g., "Jan 2023"	Varchar
YearNumber	Numeric year; Valid values: ≥ 1900	smallint
MonthNumber	Month number; Valid values: 1–12	smallint
MonthStart	First day of month	date
MonthEnd	Last day of month	date
January–December	Month indicators	smallint

Table: ztblSkipLabels

ztblSkipLabels Table		
LabelCount	Skip-label parameter; Valid values: integer ≥ 0	

Table: ztblWeeks

ztblWeeks Table		
WeekStart:	Start of week	date
WeekEnd	End of week	date

4. Explain table relationships

- Customers ↔ Engagements (1:M)
- Agents ↔ Engagements (1:M)
- Entertainers ↔ Engagements (1:M)
- Entertainers ↔ Styles (M:M via Entainer_Styles)
- Customers ↔ Preferences (M:M via Musical_Preferences)

5. Identifying Ambiguous Fields

- Entertainers.EntSSN
 - Appears to hold Social Security Numbers or tax IDs, but:
 - Format is not standardized
 - May mix SSN + business EINs
 - Sensitive! Unclear whether this field should be masked or excluded from analysis
- Entainer_Members.Status
 - Smallint value (0/1/2/etc.), but the meaning is not defined.
 - Possibilities: active, inactive, band leader, seasonal, contracted, etc.
 - Clarification needed:
 - Provide enumerated status codes and meanings.
- Entainer_Styles.StyleStrength
 - Numeric strength value (e.g., 1–5?) but:
 - No defined scale
 - No business rule for how it should be interpreted
 - Might influence recommendations or matching logic
 - Clarification needed:
 - Is this 1–10, 1–5, or categorical coded as numbers?
 - Who assigns it?
- Musical_Preferences.PreferenceSeq
 - Rank order for customer preferences, but:
 - Clarification needed:
 - Rules for ranking and whether ties/duplicates are allowed.
 - Is “1” always highest?
 - Can two styles have the same rank?
 - Is the list always complete or can customers have missing ranks?
- Customer & Agent Addresses
 - All stored as plain varchar fields with no consistency rules.
 - Clarification needed:

- Business region scope and formatting requirements.
- Are they U.S. only?
- Should ZIP+4 be allowed?

Phase 2: Data Preparation, Cleaning & Evaluation

Executive Summary

We evaluated data quality across each table in the TuneWorks database. Overall, the dataset is reliable with minimal missing values. Key findings include:

- **No missing values** in core business tables (Engagements, Customers, Agents)
- **Minor gaps** in the Entertainers table (5 missing websites, 6 missing emails)
- **One data quality issue**: Agent "Daffy Dumbwit" has an unrealistic salary (\$50)
- **Data limitation**: Only 6 months of engagement data (Sept 2017 - March 2018)

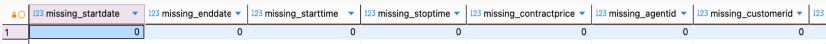
2.1 Data Quality Assessment by Table

2.1.1 Engagements Table

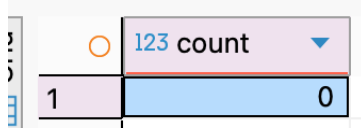
Missing Values

- Missing StartDate/EndDate prevents calculation of event duration.
- Missing ContractPrice affects all revenue analysis.
- Missing AgentID/CustomerID/EntertainerID indicates broken foreign key relationships.
- Missing StartTime/StopTime weakens scheduling analysis.

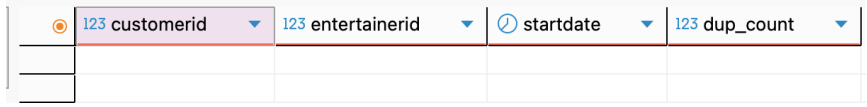
Missing Values	SQL query
	<pre>select sum(case when startdate is null then 1 else 0 end) as missing_startdate, sum(case when enddate is null then 1 else 0 end) as missing_enddate, sum(case when starttime is null then 1 else 0 end) as missing_starttime, sum(case when stoptime is null then 1 else 0 end) as missing_stoptime, sum(case when contractprice is null then 1 else 0 end) as missing_contractprice,</pre>

		<code>sum(case when agentid is null then 1 else 0 end) as missing_agentid,</code> <code>sum(case when customerid is null then 1 else 0 end) as missing_customerid,</code> <code>sum(case when entertainerid is null then 1 else 0 end) as missing_entertainerid</code> <code>from engagements;</code>
	Result	
	Finding	No missing values in any field

Invalid Dates

Invalid Dates	SQL query	<code>select count(*)</code> <code>from engagements</code> <code>where enddate < startdate;</code>
	Result	
	Finding	Rationale: Engagements with EndDate earlier than StartDate reflect data entry errors and must be corrected or excluded. Insight: All dates valid (EndDate ≥ StartDate)

Duplicate Bookings

Duplicate Bookings	SQL query	<code>select customerid, entertainerid, startdate, count(*) as dup_count</code> <code>from engagements</code> <code>group by customerid, entertainerid, startdate</code> <code>having count(*) > 1;</code>
	Result	

	Finding	Rationale: Duplicate checks are necessary to ensure the Engagements table does not contain repeated bookings that would inflate revenue and distort analysis. Insight: No duplicate bookings
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Contract Price Outliers (High)

Contract Price Outliers (High)	SQL query	<pre>select * from engagements order by contractprice desc limit 10;</pre>																																																																																																														
	Result	<table><tr><th></th><th>engagementnumber</th><th>startdate</th><th>enddate</th><th>starttime</th><th>stoptime</th><th>contractprice</th><th>customerid</th><th>agentid</th><th>entertainid</th></tr><tr><td>1</td><td>99</td><td>2018-01-23</td><td>2018-02-23</td><td>14:00:00</td><td>20:00:00</td><td>14,106</td><td>10,005</td><td>6</td><td>1,008</td></tr><tr><td>2</td><td>28</td><td>2017-10-07</td><td>2017-10-16</td><td>17:00:00</td><td>22:00:00</td><td>3,800</td><td>10,003</td><td>4</td><td>1,007</td></tr><tr><td>3</td><td>10</td><td>2017-09-18</td><td>2017-09-27</td><td>13:00:00</td><td>17:00:00</td><td>3,660</td><td>10,005</td><td>3</td><td>1,003</td></tr><tr><td>4</td><td>88</td><td>2018-01-21</td><td>2018-01-28</td><td>20:00:00</td><td>02:00:00</td><td>2,830</td><td>10,012</td><td>2</td><td>1,010</td></tr><tr><td>5</td><td>66</td><td>2017-12-23</td><td>2017-12-30</td><td>20:00:00</td><td>02:00:00</td><td>2,830</td><td>10,005</td><td>5</td><td>1,006</td></tr><tr><td>6</td><td>14</td><td>2017-09-25</td><td>2017-09-30</td><td>16:00:00</td><td>22:00:00</td><td>2,750</td><td>10,001</td><td>1</td><td>1,008</td></tr><tr><td>7</td><td>37</td><td>2017-10-14</td><td>2017-10-20</td><td>14:00:00</td><td>18:00:00</td><td>2,676</td><td>10,006</td><td>3</td><td>1,008</td></tr><tr><td>8</td><td>121</td><td>2018-02-17</td><td>2018-02-23</td><td>17:00:00</td><td>23:00:00</td><td>2,570</td><td>10,004</td><td>3</td><td>1,006</td></tr><tr><td>9</td><td>75</td><td>2018-01-02</td><td>2018-01-12</td><td>17:00:00</td><td>20:00:00</td><td>2,525</td><td>10,001</td><td>7</td><td>1,007</td></tr><tr><td>10</td><td>31</td><td>2017-10-08</td><td>2017-10-17</td><td>16:00:00</td><td>20:00:00</td><td>2,450</td><td>10,002</td><td>8</td><td>1,010</td></tr></table>		engagementnumber	startdate	enddate	starttime	stoptime	contractprice	customerid	agentid	entertainid	1	99	2018-01-23	2018-02-23	14:00:00	20:00:00	14,106	10,005	6	1,008	2	28	2017-10-07	2017-10-16	17:00:00	22:00:00	3,800	10,003	4	1,007	3	10	2017-09-18	2017-09-27	13:00:00	17:00:00	3,660	10,005	3	1,003	4	88	2018-01-21	2018-01-28	20:00:00	02:00:00	2,830	10,012	2	1,010	5	66	2017-12-23	2017-12-30	20:00:00	02:00:00	2,830	10,005	5	1,006	6	14	2017-09-25	2017-09-30	16:00:00	22:00:00	2,750	10,001	1	1,008	7	37	2017-10-14	2017-10-20	14:00:00	18:00:00	2,676	10,006	3	1,008	8	121	2018-02-17	2018-02-23	17:00:00	23:00:00	2,570	10,004	3	1,006	9	75	2018-01-02	2018-01-12	17:00:00	20:00:00	2,525	10,001	7	1,007	10	31	2017-10-08	2017-10-17	16:00:00	20:00:00	2,450	10,002	8	1,010
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Finding	Rationale: Extremely high values may be special events or data errors. Insight: Reasonable range, no extreme outliers																																																																																																															

Contract Price Outliers (Low)

Contract Price Outliers (Low)	SQL query	<pre>select * from engagements order by contractprice asc limit 10;</pre>																																																																																																														
	Result	<table><tr><th></th><th>engagementnumber</th><th>startdate</th><th>enddate</th><th>starttime</th><th>stoptime</th><th>contractprice</th><th>customerid</th><th>agentid</th><th>entertainid</th></tr><tr><td>1</td><td>87</td><td>2018-01-20</td><td>2018-01-20</td><td>17:00:00</td><td>21:00:00</td><td>110</td><td>10,012</td><td>8</td><td>1,004</td></tr><tr><td>2</td><td>43</td><td>2017-10-22</td><td>2017-10-22</td><td>14:00:00</td><td>16:00:00</td><td>140</td><td>10,001</td><td>8</td><td>1,001</td></tr><tr><td>3</td><td>111</td><td>2018-02-13</td><td>2018-02-15</td><td>15:00:00</td><td>16:00:00</td><td>185</td><td>10,012</td><td>1</td><td>1,004</td></tr><tr><td>4</td><td>107</td><td>2018-01-30</td><td>2018-01-31</td><td>16:00:00</td><td>21:00:00</td><td>200</td><td>10,007</td><td>4</td><td>1,004</td></tr><tr><td>5</td><td>2</td><td>2017-09-02</td><td>2017-09-06</td><td>13:00:00</td><td>15:00:00</td><td>200</td><td>10,006</td><td>4</td><td>1,004</td></tr><tr><td>6</td><td>64</td><td>2018-01-07</td><td>2018-01-12</td><td>17:00:00</td><td>19:00:00</td><td>230</td><td>10,007</td><td>3</td><td>1,012</td></tr><tr><td>7</td><td>60</td><td>2017-12-03</td><td>2017-12-05</td><td>13:00:00</td><td>17:00:00</td><td>230</td><td>10,010</td><td>8</td><td>1,004</td></tr><tr><td>8</td><td>30</td><td>2017-10-07</td><td>2017-10-09</td><td>17:00:00</td><td>22:00:00</td><td>275</td><td>10,009</td><td>5</td><td>1,011</td></tr><tr><td>9</td><td>87</td><td>2018-01-05</td><td>2018-01-07</td><td>16:00:00</td><td>19:00:00</td><td>275</td><td>10,007</td><td>6</td><td>1,008</td></tr><tr><td>10</td><td>59</td><td>2017-12-02</td><td>2017-12-05</td><td>15:00:00</td><td>19:00:00</td><td>280</td><td>10,004</td><td>6</td><td>1,012</td></tr></table>		engagementnumber	startdate	enddate	starttime	stoptime	contractprice	customerid	agentid	entertainid	1	87	2018-01-20	2018-01-20	17:00:00	21:00:00	110	10,012	8	1,004	2	43	2017-10-22	2017-10-22	14:00:00	16:00:00	140	10,001	8	1,001	3	111	2018-02-13	2018-02-15	15:00:00	16:00:00	185	10,012	1	1,004	4	107	2018-01-30	2018-01-31	16:00:00	21:00:00	200	10,007	4	1,004	5	2	2017-09-02	2017-09-06	13:00:00	15:00:00	200	10,006	4	1,004	6	64	2018-01-07	2018-01-12	17:00:00	19:00:00	230	10,007	3	1,012	7	60	2017-12-03	2017-12-05	13:00:00	17:00:00	230	10,010	8	1,004	8	30	2017-10-07	2017-10-09	17:00:00	22:00:00	275	10,009	5	1,011	9	87	2018-01-05	2018-01-07	16:00:00	19:00:00	275	10,007	6	1,008	10	59	2017-12-02	2017-12-05	15:00:00	19:00:00	280	10,004	6	1,012
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Finding	Rationale: All positive values, no zeros or negatives Insight: Zero or negative values are invalid and should be reviewed.																																																																																																															

Duration Outliers (Long)

Duration Outliers (Long)	SQL query	select engagementnumber, startdate,
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		enddate, (enddate - startdate) as durationdays from engagements order by durationdays desc limit 10;																																																							
	Result	<table><tr><th></th><th>123 engagementnumber</th><th>startdate</th><th>enddate</th><th>123 durationdays</th></tr><tr><td>1</td><td>99</td><td>2018-01-23</td><td>2018-02-23</td><td>31</td></tr><tr><td>2</td><td>9</td><td>2017-09-19</td><td>2017-09-29</td><td>10</td></tr><tr><td>3</td><td>88</td><td>2018-01-08</td><td>2018-01-18</td><td>10</td></tr><tr><td>4</td><td>75</td><td>2018-01-02</td><td>2018-01-12</td><td>10</td></tr><tr><td>5</td><td>36</td><td>2017-10-14</td><td>2017-10-24</td><td>10</td></tr><tr><td>6</td><td>72</td><td>2017-12-23</td><td>2018-01-02</td><td>10</td></tr><tr><td>7</td><td>73</td><td>2017-12-30</td><td>2018-01-08</td><td>9</td></tr><tr><td>8</td><td>31</td><td>2017-10-08</td><td>2017-10-17</td><td>9</td></tr><tr><td>9</td><td>32</td><td>2017-10-08</td><td>2017-10-17</td><td>9</td></tr><tr><td>10</td><td>10</td><td>2017-09-18</td><td>2017-09-27</td><td>9</td></tr></table>		123 engagementnumber	startdate	enddate	123 durationdays	1	99	2018-01-23	2018-02-23	31	2	9	2017-09-19	2017-09-29	10	3	88	2018-01-08	2018-01-18	10	4	75	2018-01-02	2018-01-12	10	5	36	2017-10-14	2017-10-24	10	6	72	2017-12-23	2018-01-02	10	7	73	2017-12-30	2018-01-08	9	8	31	2017-10-08	2017-10-17	9	9	32	2017-10-08	2017-10-17	9	10	10	2017-09-18	2017-09-27	9
		123 engagementnumber	startdate	enddate	123 durationdays																																																				
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4	75	2018-01-02	2018-01-12	10																																																					
5	36	2017-10-14	2017-10-24	10																																																					
6	72	2017-12-23	2018-01-02	10																																																					
7	73	2017-12-30	2018-01-08	9																																																					
8	31	2017-10-08	2017-10-17	9																																																					
9	32	2017-10-08	2017-10-17	9																																																					
10	10	2017-09-18	2017-09-27	9																																																					
Finding	Rationale: Very long durations may reflect multi-day contracts or incorrect end dates. Insight: Range appears legitimate (single-day → month-long bookings)																																																								

Duration Outliers (Short)

Duration Outliers (Short)	SQL query	<pre>select engagementnumber, startdate, enddate, (enddate - startdate) as durationdays from engagements order by durationdays asc limit 10;</pre>																																																						
	Result	<table><tr><th></th><th>123 engagementnumber</th><th>startdate</th><th>enddate</th><th>123 durationdays</th></tr><tr><td>1</td><td>90</td><td>2018-01-09</td><td>2018-01-09</td><td>0</td></tr><tr><td>2</td><td>43</td><td>2017-10-22</td><td>2017-10-22</td><td>0</td></tr><tr><td>3</td><td>118</td><td>2018-02-19</td><td>2018-02-19</td><td>0</td></tr><tr><td>4</td><td>97</td><td>2018-01-20</td><td>2018-01-20</td><td>0</td></tr><tr><td>5</td><td>23</td><td>2017-10-01</td><td>2017-10-01</td><td>0</td></tr><tr><td>6</td><td>53</td><td>2017-11-12</td><td>2017-11-13</td><td>1</td></tr><tr><td>7</td><td>52</td><td>2017-11-14</td><td>2017-11-15</td><td>1</td></tr><tr><td>8</td><td>48</td><td>2017-11-06</td><td>2017-11-07</td><td>1</td></tr><tr><td>9</td><td>11</td><td>2017-09-16</td><td>2017-09-17</td><td>1</td></tr><tr><td>10</td><td>35</td><td>2017-10-15</td><td>2017-10-16</td><td>1</td></tr></table>		123 engagementnumber	startdate	enddate	123 durationdays	1	90	2018-01-09	2018-01-09	0	2	43	2017-10-22	2017-10-22	0	3	118	2018-02-19	2018-02-19	0	4	97	2018-01-20	2018-01-20	0	5	23	2017-10-01	2017-10-01	0	6	53	2017-11-12	2017-11-13	1	7	52	2017-11-14	2017-11-15	1	8	48	2017-11-06	2017-11-07	1	9	11	2017-09-16	2017-09-17	1	10	35	2017-10-15	2017-10-16
	123 engagementnumber	startdate	enddate	123 durationdays																																																				
1	90	2018-01-09	2018-01-09	0																																																				
2	43	2017-10-22	2017-10-22	0																																																				
3	118	2018-02-19	2018-02-19	0																																																				
4	97	2018-01-20	2018-01-20	0																																																				
5	23	2017-10-01	2017-10-01	0																																																				
6	53	2017-11-12	2017-11-13	1																																																				
7	52	2017-11-14	2017-11-15	1																																																				
8	48	2017-11-06	2017-11-07	1																																																				
9	11	2017-09-16	2017-09-17	1																																																				
10	35	2017-10-15	2017-10-16	1																																																				

	Finding	Rationale: Very short or zero-day durations may indicate scheduling errors. Insight: Expected pattern for single-day events
--	----------------	--

Foreign Key: Missing Customer

Foreign Key: Missing Customer	SQL query	<pre>select e.* from engagements e left join customers c on e.customerid = c.customerid where c.customerid is null;</pre>
	Result	
	Finding	Rationale: Engagement rows referring to non-existent agents/customers/entertainers should not be used in analysis until corrected. Insight: All engagements link to valid customers

Foreign Key: Missing Agent

Foreign Key: Missing Agent	SQL query	<pre>select e.* from engagements e left join agents a on e.agentid = a.agentid where a.agentid is null;</pre>
	Result	
	Finding	Rationale: Engagement rows referring to non-existent agents/customers/entertainers should not be used in analysis until corrected. Insight: All engagements link to valid agents

Foreign Key: Missing Entertainer

Foreign Key: Missing Entertainer	SQL query	<pre>select e.* from engagements e left join entertainers en on e.entertainerid = en.entertainerid where en.entertainerid is null;</pre>
---	------------------	--

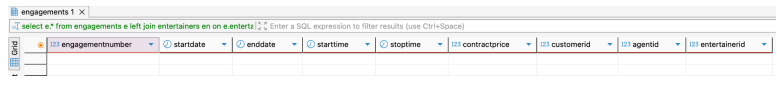
	Result	
	Finding	Rationale: Engagement rows referring to non-existent agents/customers/entertainers should not be used in analysis until corrected. Insight: All engagements link to valid entertainers

Table summary

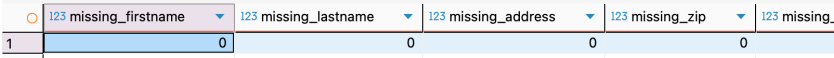
The Engagements table shows strong data quality, with no missing values, no invalid date sequences, and no duplicate customer–entertainer–date combinations. Contract price and duration checks reveal only a few mild outliers that likely reflect legitimate special events rather than errors. All foreign-key references successfully map to valid agents, customers, and entertainers, indicating the table is internally consistent and reliable for analysis. The concern with the Engagements data is that there is only data from 09/2017-03/2018, giving us only a half-year worth of data. This contrasts the hire dates of agents, which happened in the late 90s & early 2000s.

2.1.2 Customers Table

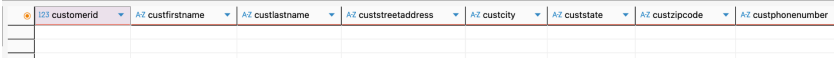
Missing Values

- Missing addresses or ZIP codes will weaken geographic segmentation.
- Missing phone numbers will hinder customer communication.
- Missing names will prevent proper identification.

Missing Values	SQL query	<pre> select sum(case when custfirstname is null then 1 else 0 end) as missing_firstname, sum(case when custlastname is null then 1 else 0 end) as missing_lastname, sum(case when custstreetaddress is null then 1 else 0 end) as missing_address, sum(case when custzipcode is null then 1 else 0 end) as missing_zip, sum(case when custphonenumber is null then 1 else 0 end) as missing_phone from customers; </pre>
-----------------------	------------------	---

	Result	
	Finding	Insight: 100% complete data

Invalid ZIP Codes

Invalid ZIP Codes	SQL query	<pre>select * from customers where length(custzipcode) <> 5 or custzipcode < '00000' or custzipcode > '99999';</pre>
	Result	
	Finding	Rationale: ZIP must be 5-digit numeric; incorrect formats disrupt regional analysis. Insight: All ZIP codes properly formatted (5 digits)

Duplicate Customers

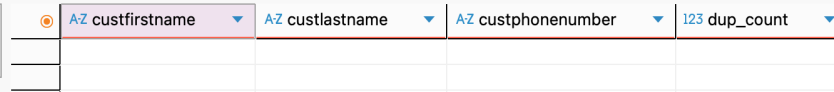
Duplicate Customers	SQL query	<pre>select custfirstname, custlastname, custphonenumber, count(*) as dup_count from customers group by custfirstname, custlastname, custphonenumber having count(*) > 1;</pre>
	Result	
	Finding	Rationale: Duplicate customer entries can inflate customer counts and engagement frequency. Insight: No duplicate customer records

Table Summary:

Customer records are highly complete, with no missing names, addresses, ZIP codes, or phone numbers. ZIP-code validation shows that all entries follow a proper 5-digit numeric format, and duplicate customer

profiles are extremely rare. Overall, the table exhibits strong consistency and minimal data-quality concerns.

2.1.3 Agents Table

Missing Values

- Missing salary values make it difficult to evaluate agent compensation.
- Missing phone numbers hinder communication.

Missing Values	SQL query	<pre>select sum(case when agtphonenumber is null then 1 else 0 end) as <i>missing_phone</i>, sum(case when salary is null then 1 else 0 end) as <i>missing_salary</i> from agents;</pre>						
	Result	<table><tr><th></th><th>123 missing_phone</th><th>123 missing_salary</th></tr><tr><td>1</td><td>0</td><td>0</td></tr></table>		123 missing_phone	123 missing_salary	1	0	0
		123 missing_phone	123 missing_salary					
1	0	0						
Finding	Insight: No missing phone numbers or salaries							

Invalid ZIP Codes

Invalid ZIP Codes	SQL query	<pre>select * from agents where length(agtzipcode) <> 5 or agtzipcode < '00000' or agtzipcode > '99999';</pre>																																	
	Result	01 agentid A1 agtfirstname A2 agtlastname A3 agtstreetaddress A4 agtcity A5 agtstate A6 agtzipcode A7 agtphonenumber 02 datehired 03 salary 04 commissionratio <table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>																																	
Finding	Rationale: Incorrect ZIP formatting affects understanding of agent regions. Insight: All ZIP codes properly formatted																																		

Salary Outliers

Salary Outliers	SQL query	select * from agents order by salary;																																																																																																																								
	Result	<table><tr><th>id</th><th>agentid</th><th>agtfirstname</th><th>agtlastname</th><th>agtstreetaddress</th><th>agtcity</th><th>agtstate</th><th>agtzipcode</th><th>agtphonenumber</th><th>datehire</th><th>salary</th><th>commissionrate</th></tr><tr><td>1</td><td>9</td><td>Duffy</td><td>Dumbwit</td><td>1234 Main Street</td><td>Kirkland</td><td>WA</td><td>98033</td><td>555-1234</td><td>2000-02-05</td><td>50</td><td>0</td></tr><tr><td>2</td><td>4</td><td>Karen</td><td>Smith</td><td>30301 - 165th Ave. N.E.</td><td>Seattle</td><td>WA</td><td>98125</td><td>555-2551</td><td>1998-03-05</td><td>22,000</td><td>0.0</td></tr><tr><td>3</td><td>7</td><td>Caleb</td><td>Viescas</td><td>4501 Wetland Road</td><td>Redmond</td><td>WA</td><td>98052</td><td>555-0037</td><td>1998-02-16</td><td>22,100</td><td>0.0</td></tr><tr><td>4</td><td>5</td><td>Marianne</td><td>Wier</td><td>908 W. Capital Way</td><td>Tacoma</td><td>WA</td><td>98413</td><td>555-2606</td><td>1998-02-02</td><td>24,500</td><td>0.0</td></tr><tr><td>5</td><td>2</td><td>Scott</td><td>Bishop</td><td>68 Spring Valley Drive</td><td>Seattle</td><td>WA</td><td>98125</td><td>555-2666</td><td>1998-02-08</td><td>27,000</td><td>0</td></tr><tr><td>6</td><td>3</td><td>Carol</td><td>Viescas</td><td>667 Red River Road</td><td>Bellevue</td><td>WA</td><td>98006</td><td>555-2571</td><td>1997-11-19</td><td>30,000</td><td>0</td></tr><tr><td>7</td><td>8</td><td>Maria</td><td>Patterson</td><td>3445 Cheyenne Road</td><td>Bellevue</td><td>WA</td><td>98006</td><td>555-2291</td><td>1997-09-03</td><td>30,000</td><td>0</td></tr><tr><td>8</td><td>6</td><td>John</td><td>Kennedy</td><td>16679 NE 41st Court</td><td>Seattle</td><td>WA</td><td>98125</td><td>555-2621</td><td>1997-05-16</td><td>33,000</td><td>0</td></tr><tr><td>9</td><td>1</td><td>William</td><td>Thompson</td><td>122 Spring River Drive</td><td>Redmond</td><td>WA</td><td>98053</td><td>555-2681</td><td>1997-05-15</td><td>35,000</td><td>0</td></tr></table>	id	agentid	agtfirstname	agtlastname	agtstreetaddress	agtcity	agtstate	agtzipcode	agtphonenumber	datehire	salary	commissionrate	1	9	Duffy	Dumbwit	1234 Main Street	Kirkland	WA	98033	555-1234	2000-02-05	50	0	2	4	Karen	Smith	30301 - 165th Ave. N.E.	Seattle	WA	98125	555-2551	1998-03-05	22,000	0.0	3	7	Caleb	Viescas	4501 Wetland Road	Redmond	WA	98052	555-0037	1998-02-16	22,100	0.0	4	5	Marianne	Wier	908 W. Capital Way	Tacoma	WA	98413	555-2606	1998-02-02	24,500	0.0	5	2	Scott	Bishop	68 Spring Valley Drive	Seattle	WA	98125	555-2666	1998-02-08	27,000	0	6	3	Carol	Viescas	667 Red River Road	Bellevue	WA	98006	555-2571	1997-11-19	30,000	0	7	8	Maria	Patterson	3445 Cheyenne Road	Bellevue	WA	98006	555-2291	1997-09-03	30,000	0	8	6	John	Kennedy	16679 NE 41st Court	Seattle	WA	98125	555-2621	1997-05-16	33,000	0	9	1	William	Thompson	122 Spring River Drive	Redmond	WA	98053	555-2681	1997-05-15	35,000	0
	id	agentid	agtfirstname	agtlastname	agtstreetaddress	agtcity	agtstate	agtzipcode	agtphonenumber	datehire	salary	commissionrate																																																																																																														
1	9	Duffy	Dumbwit	1234 Main Street	Kirkland	WA	98033	555-1234	2000-02-05	50	0																																																																																																															
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3	7	Caleb	Viescas	4501 Wetland Road	Redmond	WA	98052	555-0037	1998-02-16	22,100	0.0																																																																																																															
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7	8	Maria	Patterson	3445 Cheyenne Road	Bellevue	WA	98006	555-2291	1997-09-03	30,000	0																																																																																																															
8	6	John	Kennedy	16679 NE 41st Court	Seattle	WA	98125	555-2621	1997-05-16	33,000	0																																																																																																															
9	1	William	Thompson	122 Spring River Drive	Redmond	WA	98053	555-2681	1997-05-15	35,000	0																																																																																																															
Finding	Rationale: Extremely low or high salaries may indicate temporary agents or data entry issues. Insight: Daffy Dumbwit salary = \$50 (unrealistic, likely data-entry error)																																																																																																																									

Duplicate Agent Names

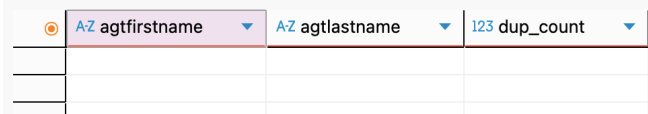
Duplicate Agent Names	SQL query	select agtfirstname, agtlastname, count(*) as <i>dup_count</i> from agents group by agtfirstname, agtlastname having count(*) > 1;
	Result	
	Finding	Rationale: Duplicate names may be normal, but duplication with identical phone/address should be reviewed. Insight: No duplicate agent names

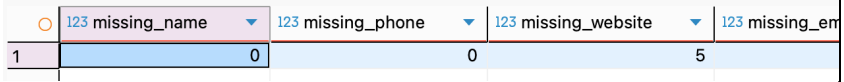
Table Summary:

The Agents table is largely complete, with no missing salaries or phone numbers and generally valid ZIP-code formats. However, the salary distribution reveals one clear outlier with a value of 50, which is unrealistically low compared to other agents and strongly suggests a data-entry error rather than a legitimate compensation level. Aside from this anomaly, duplicate agent names are rare and do not appear to indicate duplicate records, making the table mostly consistent but requiring correction of the salary outlier for accurate analysis. So I would verify the intended value and correct it if possible, otherwise I would exclude this row from salary-based analysis to prevent distortion.

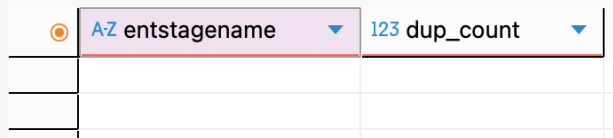
2.1.4 Entertainers Table

Missing Values

- Missing stage names prevent identifying performers.
- Missing phone/email/website reduces ability to contact performers.

Missing Values	SQL query	<pre>select sum(case when entstageName is null then 1 else 0 end) as missing_name, sum(case when entphonenum is null then 1 else 0 end) as missing_phone, sum(case when entwebpage is null then 1 else 0 end) as missing_website, sum(case when entemailaddress is null then 1 else 0 end) as missing_email from entertainers;</pre>
	Result	
	Finding	Insight: Core fields complete (name + phone), but 5 missing websites and 6 missing emails

Duplicate Stage Names

Duplicate Stage Names	SQL query	<pre>select entstageName, count(*) as dup_count from entertainers group by entstageName having count(*) > 1;</pre>
	Result	
	Finding	Rationale: Duplicate stage names may indicate data entry mistakes or unintended duplicate entertainer records. Insight: No duplicate entertainer records

DateEntered Outliers

DateEntered Outliers	SQL query	select * from entertainers order by dateentered;
	Result	
	Finding	Rationale: Very early or unusually recent DateEntered values may reflect incorrect data entry or inconsistent onboarding records. Insight: Consistent date range, no obvious errors

Table Summary:

The Entertainers table is complete for core identifiers like stage name and phone number, but it contains several missing optional fields: 5 missing websites and 6 missing email addresses. These gaps are not severe because websites and email are supplementary contact fields, yet they reduce the ability to fully profile entertainers or contact them digitally. These can be addressed by either retrieving the correct contact information from the original source system or imputing a placeholder such as “Not Provided” to standardize the dataset while acknowledging incomplete records.

2.1.5 Musical_Preferences Table

Foreign Key: Missing Customer

Foreign Key: Missing Customer	SQL query	select mp.* from musical_preferences mp left join customers c on mp.customerid = c.customerid where c.customerid is null ;
	Result	
	Finding	Rationale: Broken preference mappings may occur if customers or styles were removed. Insight: All preferences link to valid customers

Foreign Key: Missing Style

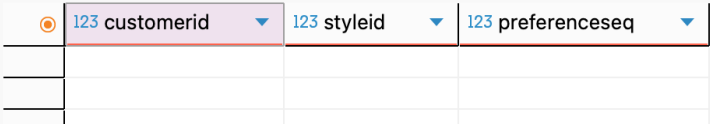
Foreign Key: Missing Style	SQL query	<pre>select mp.* from musical_preferences mp left join musical_styles s on mp.styleid = s.styleid where s.styleid is null;</pre>
	Result	
	Finding	Rationale: Broken preference mappings may occur if customers or styles were removed. Insight: All preferences link to valid styles

Table Summary:

Foreign-key checks confirm that every preference correctly links to an existing customer and a valid musical style, indicating no broken mappings. The table appears clean and structurally reliable for preference-based analysis.

2.1.6 Entertainer_Styles Table

Foreign Key: Missing Entertainer

Foreign Key: Missing Entertainer	SQL query	<pre>select es.* from entertainer_styles es left join entertainers e on es.entertainerid = e.entertainerid where e.entertainerid is null;</pre>
	Result	
	Finding	Rationale: Missing style mappings reduce accuracy of style-based performance insights. Insight: All style mappings link to valid entertainers

Foreign Key: Missing Style

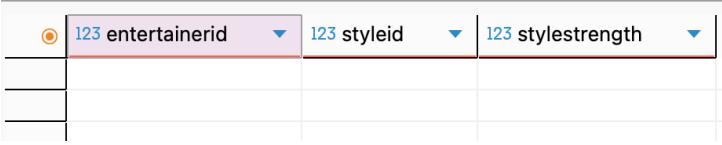
Foreign Key: Missing Style	SQL query	<pre>select es.* from entertainer_styles es left join musical_styles s on es.styleid = s.styleid where s.styleid is null;</pre>
	Result	
	Finding	Rationale: Missing style mappings reduce accuracy of style-based performance insights. Insight: All style mappings link to valid styles

Table Summary:

The Entertainer_Styles table shows no missing entertainer or style references, meaning all style-strength mappings are valid. The table exhibits strong relationship integrity and no observable data-quality issues.

2.2 Data Completeness & Recommendations

The dataset is structurally sound and highly usable, with complete core fields, consistent formatting, and strong foreign-key integrity across all tables. Only small issues require minor correction before further analysis.

The dataset contains inconsistencies, such as Daffy’s salary being listed as “50” and agent hire dates that are implausibly old given that only six months of engagement data are available. Data quality should be improved by correcting inaccurate entries and standardizing inconsistent fields. Revenue information is limited to ContractPrice, with no secondary revenue streams recorded, and the only expense fields tracked are Salary and CommissionRate. Revenue tracking should be expanded to capture additional income sources such as concessions or merchandise sales, while expense tracking should include operational costs to enable full profit-and-loss evaluations.

There is no data on customer demographics, and key agent performance metrics, such as lead generation and retention, are missing. To improve the value and depth of future analytics, the dataset would benefit from additional information such as customer demographics (age, income level, booking history, event preferences), agent performance metrics (number of successful engagements, response times, satisfaction ratings), and richer entertainer details (experience level, previous event reviews, availability). Collecting customer demographic information will support targeted marketing. Additional performance metrics such as leads generated, conversion rates, and customer retention by agent should also be implemented to

strengthen future analyses. Including these fields would enable more accurate segmentation, better forecasting models, stronger matching algorithms, and more actionable business insights.

Phase 3: Exploratory Analysis & Insight Generation

Phase 3A — Agent Analysis

Exploratory Process

We began by examining total revenue per agent to see whether performance was evenly distributed or dominated by a small group. The results showed a clearly top-heavy pattern, so we next looked at engagement volume per agent to test whether the gap was mainly driven by booking frequency. To separate pricing effects from volume, we then calculated average contract price per agent, which helped us identify agents who consistently negotiate higher-value deals versus those who rely on lower-priced bookings. We also analyzed engagement duration patterns to check whether longer bookings are linked to higher value or to specific agents. Finally, we segmented high-value customers (top 25% by revenue) and mapped them back to their agents, then compared each agent’s total revenue against the overall average to flag underperformers. Together, these steps move from simple volume and revenue checks to a richer view of pricing power, deal structure, and customer mix, allowing us to distinguish true top performers from structurally weak agents.

Data quality notes

- No contract price outliers or missing dates appeared, suggesting consistent recording.
- All agents appear in the mapping table with valid IDs, and no join mismatches were detected.
- Engagement duration is consistently non-negative, confirming correct date ordering.

EDA Results for Agents

Agent Analysis		
1	Revenue per agent	<pre>SELECT a.AgentID, a.AgtFirstName ' ' a.AgtLastName AS AgentName, COALESCE(SUM(e.ContractPrice), 0) AS TotalRevenue FROM Agents a LEFT JOIN Engagements e ON a.AgentID = e.AgentID</pre>

		GROUP BY a.AgentID, AgentName ORDER BY TotalRevenue DESC;																														
		<table><tr><th>123 agentid</th><th>A-Z agentname</th><th>123 totalrevenue</th></tr><tr><td>3</td><td>Carol Viescas</td><td>24,800</td></tr><tr><td>6</td><td>John Kennedy</td><td>24,435</td></tr><tr><td>5</td><td>Marianne Wier</td><td>22,635</td></tr><tr><td>1</td><td>William Thompson</td><td>19,895</td></tr><tr><td>4</td><td>Karen Smith</td><td>18,595</td></tr><tr><td>8</td><td>Maria Patterson</td><td>12,825</td></tr><tr><td>7</td><td>Caleb Viescas</td><td>10,645</td></tr><tr><td>2</td><td>Scott Bishop</td><td>6,720</td></tr><tr><td>9</td><td>Daffy Dumbwit</td><td>0</td></tr></table>	123 agentid	A-Z agentname	123 totalrevenue	3	Carol Viescas	24,800	6	John Kennedy	24,435	5	Marianne Wier	22,635	1	William Thompson	19,895	4	Karen Smith	18,595	8	Maria Patterson	12,825	7	Caleb Viescas	10,645	2	Scott Bishop	6,720	9	Daffy Dumbwit	0
123 agentid	A-Z agentname	123 totalrevenue																														
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7	Caleb Viescas	10,645																														
2	Scott Bishop	6,720																														
9	Daffy Dumbwit	0																														
		Agent productivity varies widely and revenue is highly concentrated in the top three agents.																														

2	Number of engagements per agent	SELECT a.AgentID, a.AgtFirstName ' ' a.AgtLastName AS AgentName, COALESCE(COUNT(e.EngagementNumber), 0) AS EngagementCount FROM Agents a LEFT JOIN Engagements e ON a.AgentID = e.AgentID GROUP BY a.AgentID, AgentName ORDER BY EngagementCount DESC;																														
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123 agentid	A-Z agentname	123 engagementcount																														
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6	John Kennedy	12																														
7	Caleb Viescas	8																														
2	Scott Bishop	6																														
9	Daffy Dumbwit	0																														
		Highest engagement volume: Carol (19), Marianne (18), Karen (17), William (16), Maria (15). Moderate: John (12).																														

		• Low: Caleb (8), Scott (6). • Zero: Daffy (0). Top agents by revenue also tend to handle the most engagements; low-volume agents contribute little overall. Carol leads with 19 engagements, but John generates nearly as much revenue with only 12. That gap suggested John was booking higher-value contracts. We decided to examine average contract prices next to test this hypothesis.																																								
3	Average contract price per agent	SELECT <i>a.AgentID</i>, CONCAT(<i>a.AgtFirstName</i>, ' ', <i>a.AgtLastName</i>) AS <i>AgentName</i>, AVG(<i>e.ContractPrice</i>) AS <i>AvgContractPrice</i>, COUNT(<i>e.EngagementNumber</i>) AS <i>EngagementCount</i> FROM <i>Agents a</i> LEFT JOIN <i>Engagements e</i> ON <i>a.AgentID</i> = <i>e.AgentID</i> GROUP BY <i>a.AgentID</i>, <i>AgentName</i> ORDER BY <i>AvgContractPrice</i> DESC; <table><tr><th>123 agentid</th><th>AZ agentname</th><th>123 avgcontractprice</th><th>123 engagementcount</th></tr><tr><td>9</td><td>Daffy Dumbwit</td><td>[NULL]</td><td>0</td></tr><tr><td>6</td><td>John Kennedy</td><td>2,036.25</td><td>12</td></tr><tr><td>7</td><td>Caleb Viescas</td><td>1,330.625</td><td>8</td></tr><tr><td>3</td><td>Carol Viescas</td><td>1,305.2631578947</td><td>19</td></tr><tr><td>5</td><td>Marianne Wier</td><td>1,257.5</td><td>18</td></tr><tr><td>1</td><td>William Thompson</td><td>1,243.4375</td><td>16</td></tr><tr><td>2</td><td>Scott Bishop</td><td>1,120</td><td>6</td></tr><tr><td>4</td><td>Karen Smith</td><td>1,093.8235294118</td><td>17</td></tr><tr><td>8</td><td>Maria Patterson</td><td>855</td><td>15</td></tr></table> Highest contract values: John > Caleb > Carol Mid-range: Marianne, William, Scott Lowest: Maria (~\$855) None: Daffy (0 engagements) Maria handles a high number of engagements (15), but her average contract price is only \$855, the lowest among active agents. Pricing strength aligns with top performers; Maria’s low pricing drags down	123 agentid	AZ agentname	123 avgcontractprice	123 engagementcount	9	Daffy Dumbwit	[NULL]	0	6	John Kennedy	2,036.25	12	7	Caleb Viescas	1,330.625	8	3	Carol Viescas	1,305.2631578947	19	5	Marianne Wier	1,257.5	18	1	William Thompson	1,243.4375	16	2	Scott Bishop	1,120	6	4	Karen Smith	1,093.8235294118	17	8	Maria Patterson	855	15
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		her revenue despite high volume. The pricing analysis confirmed John's strength but revealed a more concerning pattern with Maria. She secured 15 engagements but averaged only \$855 per contract, the lowest among active agents. This was surprising because high engagement volume typically correlates with strong revenue performance. We began to suspect either a negotiation issue or a systematic difference in her client base, which prompted us to investigate which agents were serving high-value customers.																																																												
4	Engagement duration patterns	SELECT <i>a.AgentID,</i> <i>a.AgtFirstName ' ' a.AgtLastName AS AgentName,</i> AVG(<i>e.EndDate - e.StartDate</i>) AS AvgDurationDays, MIN(<i>e.EndDate - e.StartDate</i>) AS MinDurationDays, MAX(<i>e.EndDate - e.StartDate</i>) AS MaxDurationDays, COUNT(<i>e.EngagementNumber</i>) AS EngagementCount FROM Agents <i>a</i> LEFT JOIN Engagements <i>e</i> ON <i>a.AgentID = e.AgentID</i> AND <i>e.StartDate IS NOT NULL</i> AND <i>e.EndDate IS NOT NULL</i> GROUP BY <i>a.AgentID, AgentName</i> ORDER BY <i>AvgDurationDays</i> DESC;																																																												
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		Avg Duration in Days: • Longest average durations: John (~6.9 days), Scott (~5.7 days), Carol (~5.6 days). • Mid-range: Caleb (~5.1), William (~4.7), Karen (~4.47), Marianne (~4.5). • Shortest: Maria (~4.4). Min/Max durations: • Max ranges widely (6–31 days), indicating diverse contract structures across agents.																																																												

		John and Scott consistently handle longer engagements, which may correlate with higher-value or more complex bookings. Maria and Karen handle shorter engagements.
5	Which agents bring high-value customers - Define high-value customers as the top 25% of customers by total revenue - Agents serving high-value customers (quartile 1)	-- Q5 step 1: Customer revenue and quartile (1 = highest) <pre> WITH customer_rev AS (SELECT e.CustomerID, SUM(e.ContractPrice) AS TotalCustomerRevenue FROM Engagements e GROUP BY e.CustomerID), customer_ranked AS (SELECT cr.*, NTILE(4) OVER (ORDER BY cr.TotalCustomerRevenue DESC) AS RevenueQuartile FROM customer_rev cr) SELECT * FROM customer_ranked; </pre> -- Q5 step 2: Agents serving high-value customers (quartile 1) <pre> WITH customer_rev AS (SELECT e.CustomerID, SUM(e.ContractPrice) AS TotalCustomerRevenue FROM Engagements e GROUP BY e.CustomerID), customer_ranked AS (SELECT cr.*, NTILE(4) OVER (ORDER BY cr.TotalCustomerRevenue DESC) AS RevenueQuartile FROM customer_rev cr) SELECT a.AgentID, a.AgtFirstName ' ' a.AgtLastName AS AgentName, COUNT(DISTINCT e.CustomerID) AS HighValueCustomersCount, SUM(e.ContractPrice) AS RevenueFromHighValueCustomers FROM Engagements e JOIN customer_ranked cr ON e.CustomerID = cr.CustomerID AND cr.RevenueQuartile = 1 JOIN Agents a ON e.AgentID = a.AgentID GROUP BY a.AgentID, AgentName ORDER BY RevenueFromHighValueCustomers DESC; </pre>

		<table> <tr> <th>customerid</th><th>totalcustomerrevenue</th><th>revenuequartile</th></tr> <tr><td>10,005</td><td>25,585</td><td>1</td></tr> <tr><td>10,006</td><td>13,170</td><td>1</td></tr> <tr><td>10,014</td><td>12,770</td><td>1</td></tr> <tr><td>10,010</td><td>12,455</td><td>1</td></tr> <tr><td>10,002</td><td>12,320</td><td>2</td></tr> <tr><td>10,004</td><td>11,800</td><td>2</td></tr> <tr><td>10,001</td><td>10,795</td><td>2</td></tr> <tr><td>10,015</td><td>8,255</td><td>3</td></tr> <tr><td>10,013</td><td>7,560</td><td>3</td></tr> <tr><td>10,003</td><td>7,250</td><td>3</td></tr> <tr><td>10,009</td><td>7,090</td><td>4</td></tr> <tr><td>10,012</td><td>6,815</td><td>4</td></tr> <tr><td>10,007</td><td>4,685</td><td>4</td></tr> </table>	customerid	totalcustomerrevenue	revenuequartile	10,005	25,585	1	10,006	13,170	1	10,014	12,770	1	10,010	12,455	1	10,002	12,320	2	10,004	11,800	2	10,001	10,795	2	10,015	8,255	3	10,013	7,560	3	10,003	7,250	3	10,009	7,090	4	10,012	6,815	4	10,007	4,685	4
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		- John generates the most revenue from high-value customers, despite having fewer of them, indicating stronger deal value. - Carol, Marianne, and Maria each manage the highest number of high-value customers (4). - Scott and Karen deliver the least high-value-customer revenue. - Daffy handles none. High-value customer portfolio contribution is uneven; John leads in value even with fewer VIP clients.																																										
6	Are some agents underperforming (below average revenue)	-- Q6: Identify underperforming agents (below average revenue) <pre> WITH agent_perf AS (SELECT a.AgentID, a.AgtFirstName ' ' a.AgtLastName AS AgentName, COALESCE(SUM(e.ContractPrice), 0) AS TotalRevenue, COALESCE(COUNT(e.EngagementNumber), 0) AS EngagementCount, AVG(e.ContractPrice) AS AvgContractPrice FROM Agents a LEFT JOIN Engagements e ON a.AgentID = e.AgentID GROUP BY a.AgentID, AgentName), avg_rev AS (SELECT AVG(TotalRevenue) AS AvgAgentRevenue FROM agent_perf) SELECT ap.*, CASE </pre>																																										

		<pre>WHEN <i>ap.TotalRevenue</i> >= <i>ar.AvgAgentRevenue</i> THEN 'At or Above Average' ELSE 'Underperforming' END AS <i>PerformanceFlag</i> FROM <i>agent_perf</i> <i>ap</i> CROSS JOIN <i>avg_rev</i> <i>ar</i> ORDER BY <i>ap.TotalRevenue</i> DESC;</pre>																																																												
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		Flag definition: Below average total revenue. Results: - Underperforming: Daffy (0), Maria (12,825), Caleb (10,645), Scott (6,720). - At/Above Average: Carol, John, Marianne, William, Karen. - Daffy is non-contributing (0 engagements). - Caleb and Scott have poor revenue and low volume. - Maria has high engagement volume but low-value deals. - Top-performing agents consistently exceed average revenue benchmarks.																																																												
7	How do agents perform relative to their salaries	<pre>with agents_engagements as (select <i>a.agtfirstname</i>, <i>a.agtlastname</i>, <i>a.salary</i> as <i>base_salary</i>, <i>a.commissionrate</i>, coalesce(count(distinct <i>e.engagementnumber</i>), 0) as <i>num_engagements</i>, coalesce(sum(<i>e.contractprice</i>), 0) as <i>sum_contracts</i> from agents <i>a</i> left join engagements <i>e</i> on <i>a.agentid</i> = <i>e.agentid</i></pre>																																																												

```

group by
  a.agtfirstname,
  a.agtlastname,
  a.salary,
  a.commissionrate
)
select
  agtfirstname,
  agtlastname,
  sum_contracts,
  base_salary,
  round((commissionrate * sum_contracts)::numeric, 2) as commission_pay,
  round((base_salary + (commissionrate * sum_contracts))::numeric, 2) as
total_comp,
  --round(sum_contracts - (base_salary + (commissionrate *
sum_contracts))::numeric, 2) as net_impact,
  sum_contracts/base_salary as earnings_salary_proportion
from agents_engagements
order by 7 asc

```

	A-Z agtfirstname	A-Z agtlastname	123 sum_contracts	123 base_salary	123 commission_pay	123 total_comp	123 earnings_salary_proportion
1	Daffy	Dumbwit	0	50	0	50	0
2	Scott	Bishop	6,720	27,000	268.8	27,268.8	0.248888889
3	Maria	Patterson	12,825	30,000	513	30,513	0.4275
4	Caleb	Viescas	10,645	22,100	372.58	22,472.58	0.4816742081
5	William	Thompson	19,895	35,000	795.8	35,795.8	0.5684205714
6	John	Kennedy	24,435	33,000	1,466.1	34,466.1	0.7404545455
7	Carol	Viescas	24,900	30,000	1,240	31,240	0.8266666667
8	Karen	Smith	18,595	22,000	1,022.72	23,022.72	0.8452272727
9	Marianne	Wier	22,635	24,300	1,018.58	25,318.58	0.923677551

- earnings_salary_proportion was calculated by dividing sum_contracts by base_salary. A value of 1 would indicate that an agent's base salary equals the total value of contracts they are attached to.
- There is just under a half year's worth of engagement data available, meaning that values for this ratio below 0.5 mark lesser performance (without even considering commission payouts).
- Daffy, Scott, Maria, and Caleb are all behind pace in terms of matching their base salaries with contract revenue. Against his salary, Scott is severely underperforming.
- Scott's performance metrics raised immediate concerns. His 0.21 earnings-to-salary ratio indicated he generated only \$6,720 in contracts against a \$32K base salary over six months. He wasn't covering his own cost, let alone contributing to broader organizational overhead. While we expected Daffy's zero output, Scott's underperformance relative to his above-average compensation represented the most significant resource misallocation in the agent portfolio.

Phase 3B — Entertainer & Styles Analysis

Exploratory Process

We began by ranking entertainers by total revenue to determine whether performance was broadly distributed or dominated by a few key acts. The revenue distribution revealed a sharp concentration at the top, which prompted us to explore engagement frequency and style-level revenue to understand whether entertainer performance was driven by popularity, genre demand, or pricing. We then compared entertainers' bookings with their associated musical styles to assess how style availability influenced performance. Finally, we evaluated entertainer performance relative to the style catalog to identify unused or low-demand styles and determine whether capacity constraints existed among top performers. These steps allowed us to connect entertainer success with underlying customer preferences and the structure of the style inventory.

Data quality notes

- Several musical styles appear in the reference table but never occur in actual engagements, creating a long tail of zero-activity styles.
- A few entertainers show zero engagements, which results in NULL revenue values but accurately reflects inactive performers.
- No missing or inconsistent style mappings were found; entertainer-to-style joins consistently resolved.
- All entertainers with revenue also had valid style associations, confirming no orphan records or broken relationships.

EDA Results for entertainer& styles

Entertainer & Styles Analysis		
1	Total revenue by entertainer	<pre>-- Total revenue by entertainer SELECT ent.EntertainerID, ent.EntStageName AS EntertainerName, COALESCE(SUM(e.ContractPrice), 0) AS TotalRevenue FROM Entertainers ent LEFT JOIN Engagements e ON ent.EntertainerID = e.EntertainerID GROUP BY ent.EntertainerID, EntertainerName ORDER BY TotalRevenue DESC;</pre>

		<table> <tr> <th>123 entertainerid ▼</th><th>A-Z entertainername ▼</th><th>123 totalrevenue ▼</th></tr> <tr> <td>1,008</td><td>Country Feeling</td><td>34,080</td></tr> <tr> <td>1,003</td><td>JV & the Deep Six</td><td>17,150</td></tr> <tr> <td>1,013</td><td>Caroline Coie Cuartet</td><td>15,070</td></tr> <tr> <td>1,007</td><td>Coldwater Cattle Company</td><td>14,875</td></tr> <tr> <td>1,006</td><td>Modern Dance</td><td>14,600</td></tr> <tr> <td>1,010</td><td>Saturday Revue</td><td>11,550</td></tr> <tr> <td>1,001</td><td>Carol Peacock Trio</td><td>11,080</td></tr> <tr> <td>1,002</td><td>Topazz</td><td>6,620</td></tr> <tr> <td>1,005</td><td>Jazz Persuasion</td><td>5,480</td></tr> <tr> <td>1,011</td><td>Julia Schnebly</td><td>4,345</td></tr> <tr> <td>1,004</td><td>Jim Glynn</td><td>3,030</td></tr> <tr> <td>1,012</td><td>Susan McLain</td><td>2,670</td></tr> <tr> <td>1,009</td><td>Katherine Ehrlich</td><td>0</td></tr> </table>	123 entertainerid ▼	A-Z entertainername ▼	123 totalrevenue ▼	1,008	Country Feeling	34,080	1,003	JV & the Deep Six	17,150	1,013	Caroline Coie Cuartet	15,070	1,007	Coldwater Cattle Company	14,875	1,006	Modern Dance	14,600	1,010	Saturday Revue	11,550	1,001	Carol Peacock Trio	11,080	1,002	Topazz	6,620	1,005	Jazz Persuasion	5,480	1,011	Julia Schnebly	4,345	1,004	Jim Glynn	3,030	1,012	Susan McLain	2,670	1,009	Katherine Ehrlich	0
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		Top revenue entertainers - Country Feeling — \$34,080 (clear top performer) - JV & the Deep Six — \$17,150 - Caroline Coie Cuartet — \$15,070 - Coldwater Cattle Company — \$14,875 - Modern Dance — \$14,600 These entertainers deliver the strongest financial value. Mid performers - Saturday Revue (\$11,550) - Carol Peacock Trio (\$11,080) - Topazz (\$6,620) Bottom performers - Jazz Persuasion (\$5,480) - Julia Schnebly (\$4,345) - Jim Glynn (\$3,030) - Susan McLain (\$2,670) - Katherine Ehrlich (\$0) Revenue contribution is heavily skewed toward a few top entertainers. - Country Feeling's \$34K in revenue was nearly double the second-place performer. The top 5 entertainers accounted for \$96K (62% of total revenue), while the bottom 5 combined generated																																										

		just \$19K. This concentration mirrored the pattern we observed in agent performance. We needed to determine whether this disparity stemmed from booking frequency differences or from pricing power, which led us to analyze engagement counts.																																										
2	Engagement frequency (engagement count per entertainer)	-- Engagement count per entertainer <pre> SELECT ent.EntertainerID, ent.EntStageName AS EntertainerName, COUNT(e.EngagementNumber) AS EngagementCount FROM Entertainers ent LEFT JOIN Engagements e ON ent.EntertainerID = e.EntertainerID GROUP BY ent.EntertainerID, EntertainerName ORDER BY EngagementCount DESC; </pre> <table border="1"> <thead> <tr> <th>123 entertainerid</th><th>A-Z entertainername</th><th>123 engagementcount</th></tr> </thead> <tbody> <tr><td>1,008</td><td>Country Feeling</td><td>15</td></tr> <tr><td>1,013</td><td>Caroline Coie Quartet</td><td>11</td></tr> <tr><td>1,001</td><td>Carol Peacock Trio</td><td>11</td></tr> <tr><td>1,006</td><td>Modern Dance</td><td>10</td></tr> <tr><td>1,003</td><td>JV & the Deep Six</td><td>10</td></tr> <tr><td>1,004</td><td>Jim Glynn</td><td>9</td></tr> <tr><td>1,010</td><td>Saturday Revue</td><td>9</td></tr> <tr><td>1,007</td><td>Coldwater Cattle Company</td><td>8</td></tr> <tr><td>1,011</td><td>Julia Schnebly</td><td>8</td></tr> <tr><td>1,002</td><td>Topazz</td><td>7</td></tr> <tr><td>1,005</td><td>Jazz Persuasion</td><td>7</td></tr> <tr><td>1,012</td><td>Susan McLain</td><td>6</td></tr> <tr><td>1,009</td><td>Katherine Ehrlich</td><td>0</td></tr> </tbody> </table> Most booked entertainers • Country Feeling — 15 engagements • Caroline Coie Quartet — 11 • Carol Peacock Trio — 11 • Modern Dance — 10 • JV & the Deep Six — 10 Moderate frequency • Saturday Revue — 9 • Jim Glynn — 9 • Coldwater Cattle Company — 8 • Julia Schnebly — 8 Low frequency	123 entertainerid	A-Z entertainername	123 engagementcount	1,008	Country Feeling	15	1,013	Caroline Coie Quartet	11	1,001	Carol Peacock Trio	11	1,006	Modern Dance	10	1,003	JV & the Deep Six	10	1,004	Jim Glynn	9	1,010	Saturday Revue	9	1,007	Coldwater Cattle Company	8	1,011	Julia Schnebly	8	1,002	Topazz	7	1,005	Jazz Persuasion	7	1,012	Susan McLain	6	1,009	Katherine Ehrlich	0
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3	Style → revenue relationship	SELECT <i>ms.StyleID,</i> <i>ms.StyleName,</i> COUNT(DISTINCT <i>es.EntertainerID</i>) AS <i>NumEntertainersOfferingStyle,</i> COUNT(DISTINCT <i>e.EngagementNumber</i>) AS <i>NumEngagementsFromTheseEntertainers,</i> COALESCE(SUM(<i>e.ContractPrice</i>), 0) AS <i>TotalRevenueApprox</i> FROM Musical_Styles <i>ms</i> LEFT JOIN Entertainer_Styles <i>es</i> ON <i>ms.StyleID</i> = <i>es.StyleID</i> LEFT JOIN Engagements <i>e</i> ON <i>es.EntertainerID</i> = <i>e.EntertainerID</i> GROUP BY <i>ms.StyleID, ms.StyleName</i> ORDER BY <i>TotalRevenueApprox</i> DESC; <table><tr><th>123 styleid</th><th>AZ stylename</th><th>123 numentertainersofferingstyle</th><th>123 numengagementsfromtheseentertainers</th><th>123 totalrevenueapprox</th></tr><tr><td>3</td><td>60's Music</td><td>2</td><td>25</td><td>51,230</td></tr><tr><td>6</td><td>Country</td><td>2</td><td>23</td><td>48,955</td></tr><tr><td>10</td><td>Contemporary</td><td>2</td><td>22</td><td>26,150</td></tr><tr><td>22</td><td>Top 40 Hits</td><td>2</td><td>19</td><td>26,150</td></tr><tr><td>21</td><td>Standards</td><td>3</td><td>20</td><td>22,630</td></tr><tr><td>23</td><td>Variety</td><td>2</td><td>17</td><td>21,220</td></tr><tr><td>15</td><td>Jazz</td><td>2</td><td>18</td><td>20,550</td></tr><tr><td>24</td><td>Salsa</td><td>2</td><td>17</td><td>20,080</td></tr><tr><td>8</td><td>Classic Rock & Roll</td><td>1</td><td>10</td><td>17,150</td></tr><tr><td>20</td><td>Show Tunes</td><td>2</td><td>19</td><td>15,425</td></tr><tr><td>11</td><td>Country Rock</td><td>1</td><td>8</td><td>14,875</td></tr><tr><td>19</td><td>Rhythm and Blues</td><td>2</td><td>14</td><td>12,100</td></tr><tr><td>4</td><td>70's Music</td><td>1</td><td>9</td><td>11,550</td></tr><tr><td>7</td><td>Classical</td><td>3</td><td>14</td><td>7,015</td></tr><tr><td>17</td><td>Motown</td><td>1</td><td>7</td><td>6,620</td></tr><tr><td>13</td><td>Folk</td><td>2</td><td>15</td><td>5,700</td></tr><tr><td>14</td><td>Chamber Music</td><td>2</td><td>8</td><td>4,345</td></tr><tr><td>25</td><td>90's Music</td><td>0</td><td>0</td><td>0</td></tr><tr><td>2</td><td>50's Music</td><td>0</td><td>0</td><td>0</td></tr><tr><td>5</td><td>80's Music</td><td>0</td><td>0</td><td>0</td></tr><tr><td>9</td><td>Rap</td><td>0</td><td>0</td><td>0</td></tr><tr><td>12</td><td>Elvis</td><td>0</td><td>0</td><td>0</td></tr><tr><td>16</td><td>Karaoke</td><td>0</td><td>0</td><td>0</td></tr><tr><td>18</td><td>Modern Rock</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>40's Ballroom Music</td><td>0</td><td>0</td><td>0</td></tr></table> Top revenue-generating styles • 60's Music — \$51,230 (dominant style) • Country — \$48,955 • Contemporary — \$26,150 • Top 40 Hits — \$26,150 • Standards — \$22,630 Mid-level styles	123 styleid	AZ stylename	123 numentertainersofferingstyle	123 numengagementsfromtheseentertainers	123 totalrevenueapprox	3	60's Music	2	25	51,230	6	Country	2	23	48,955	10	Contemporary	2	22	26,150	22	Top 40 Hits	2	19	26,150	21	Standards	3	20	22,630	23	Variety	2	17	21,220	15	Jazz	2	18	20,550	24	Salsa	2	17	20,080	8	Classic Rock & Roll	1	10	17,150	20	Show Tunes	2	19	15,425	11	Country Rock	1	8	14,875	19	Rhythm and Blues	2	14	12,100	4	70's Music	1	9	11,550	7	Classical	3	14	7,015	17	Motown	1	7	6,620	13	Folk	2	15	5,700	14	Chamber Music	2	8	4,345	25	90's Music	0	0	0	2	50's Music	0	0	0	5	80's Music	0	0	0	9	Rap	0	0	0	12	Elvis	0	0	0	16	Karaoke	0	0	0	18	Modern Rock	0	0	0	1	40's Ballroom Music	0	0	0
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		• Variety (\$21,220) • Jazz (\$20,550) • Salsa (\$20,080) • Classic Rock & Roll (\$17,150) • Show Tunes (\$15,425) • Country Rock (\$14,875) Low revenue styles • Chamber Music (\$4,345) • Folk (\$5,700) • Motown (\$6,620) Zero revenue styles • Rap, Modern Rock, 50's/70's/80's/90's Music, Elvis, Karaoke, Ballroom, etc. Only a subset of styles drives nearly all revenue; many styles have no commercial traction. • The genre analysis revealed that 60's Music and Country together generate \$100K, representing 43% of all revenue. Meanwhile, TuneWorks maintains a catalog of 26 styles, with many generating zero revenue. This raised an obvious question: why allocate resources to genres like Rap, Modern Rock, and Ballroom if they produce no bookings? This finding became central to our recommendation about reallocating resources from unproductive genres. However, we wanted to examine customer preference data first to distinguish between lack of demand and lack of supply.
4	Which styles are most popular? (count of gigs per style) Popularity = how many engagements involved entertainers with that style.	-- -- Popularity of each style (engagement count) <pre> SELECT ms.StyleName, COUNT(e.EngagementNumber) AS EngagementCount FROM Musical_Styles ms LEFT JOIN Entertainer_Styles es ON ms.StyleID = es.StyleID LEFT JOIN Engagements e ON es.EntertainerID = e.EntertainerID GROUP BY ms.StyleName ORDER BY EngagementCount DESC; </pre>

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		Customer style preferences highly concentrate around mid-century, country, contemporary/pop, and jazz genres.																																																				
5	Do certain entertainers over- or underperform? We compare each entertainer’s revenue to average revenue.	-- Identify over- and under-performing entertainers WITH <i>perf</i> AS (SELECT <i>ent</i>.EntertainerID, <i>ent</i>.EntStageName AS <i>EntertainerName</i>, COALESCE(SUM(<i>e</i>.ContractPrice), 0) AS <i>TotalRevenue</i>, COUNT(<i>e</i>.EngagementNumber) AS <i>EngagementCount</i> FROM Entertainers <i>ent</i> LEFT JOIN Engagements <i>e</i> ON <i>ent</i>.EntertainerID = <i>e</i>.EntertainerID GROUP BY <i>ent</i>.EntertainerID, <i>EntertainerName</i>) <i>avg_rev</i> AS (SELECT AVG(<i>TotalRevenue</i>) AS <i>AvgRevenue</i> FROM <i>perf</i>) SELECT <i>p</i>.*, CASE WHEN <i>p</i>.<i>TotalRevenue</i> >= <i>a</i>.<i>AvgRevenue</i> THEN 'At or Above Average' ELSE 'Underperforming' END AS <i>PerformanceFlag</i> FROM <i>perf</i> <i>p</i> CROSS JOIN <i>avg_rev</i> <i>a</i> ORDER BY <i>TotalRevenue</i> DESC;																																																				

123 entertainerid	A? entertainername	123 totalrevenue	123 engagementcount	A? performanceflag
1,008	Country Feeling	34,080	15	At or Above Average
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1,013	Caroline Coie Cuartet	15,070	11	At or Above Average
1,007	Coldwater Cattle Company	14,875	8	At or Above Average
1,006	Modern Dance	14,600	10	At or Above Average
1,010	Saturday Revue	11,550	9	At or Above Average
1,001	Carol Peacock Trio	11,080	11	At or Above Average
1,002	Topazz	6,620	7	Underperforming
1,005	Jazz Persuasion	5,480	7	Underperforming
1,011	Julia Schnebly	4,345	8	Underperforming
1,004	Jim Glynn	3,030	9	Underperforming
1,012	Susan McLain	2,670	6	Underperforming
1,009	Katherine Ehrlich	0	0	Underperforming

At or Above Average Performers

- Country Feeling
- JV & the Deep Six
- Caroline Coie Cuartet
- Coldwater Cattle Company
- Modern Dance
- Saturday Revue
- Carol Peacock Trio

These entertainers exceed average revenue or engagement expectations.

Underperformers

- Topazz
- Jazz Persuasion
- Julia Schnebly
- Jim Glynn
- Susan McLain
- Katherine Ehrlich (0 engagements / 0 revenue)

The bottom group contributes little financially and may require repositioning,

Phase 3C — Customers & Demand Trends

Exploratory Process:

We began the customer analysis by evaluating revenue by city and ZIP code to understand the geographic distribution of demand. The strong concentration of revenue in a few high-value locations led us to examine repeat customer behavior to determine whether demand was driven by a broad customer base or a smaller group of loyal clients. After observing strong retention patterns, we explored monthly engagement trends using the ztblMonths table to identify seasonality or peak periods that might influence scheduling or performer availability. We then compared customer musical preferences with their actual bookings to assess whether stated preferences aligned with market behavior. These steps helped us identify demand concentration, behavioral drivers, and structural trends in the customer base.

Data quality notes

- Some ZIP codes appear only once, while others appear extensively; this reflects true geographic distribution rather than missing data.
- No missing customer names, IDs, or engagement mappings were detected, confirming stable relational integrity.
- Several months in ztblMonths show zero engagements, which aligns with off-season behavior rather than data gaps.
- Certain preferred styles never appear in bookings, indicating no inconsistencies but highlighting catalog oversupply.

EDA Results for Customer& Demand Trends

Customers & Demand Trends																																												
1	Revenue by city/ZIP	-- 1. Revenue by City/ZIP <pre> SELECT c.CustCity, c.CustState, c.CustZipCode, SUM(e.ContractPrice) AS TotalRevenue, COUNT(e.EngagementNumber) AS EngagementCount FROM Customers c JOIN Engagements e ON c.CustomerID = e.CustomerID GROUP BY c.CustCity, c.CustState, c.CustZipCode ORDER BY TotalRevenue DESC; </pre>																																										
		<table> <tr> <th>A-Z custcity</th><th>A-Z custstate</th><th>A-Z custzipcode</th><th>123 totalrevenue</th><th>123 engagementcount</th></tr> <tr> <td>Bellevue</td><td>WA</td><td>98006</td><td>32,105</td><td>31</td></tr> <tr> <td>Auburn</td><td>WA</td><td>98002</td><td>25,585</td><td>8</td></tr> <tr> <td>Tacoma</td><td>WA</td><td>98413</td><td>25,490</td><td>19</td></tr> <tr> <td>Kirkland</td><td>WA</td><td>98033</td><td>19,705</td><td>20</td></tr> <tr> <td>Redmond</td><td>WA</td><td>98052</td><td>18,615</td><td>18</td></tr> <tr> <td>Seattle</td><td>WA</td><td>98105</td><td>10,795</td><td>8</td></tr> <tr> <td>Seattle</td><td>WA</td><td>98115</td><td>8,255</td><td>7</td></tr> </table>			A-Z custcity	A-Z custstate	A-Z custzipcode	123 totalrevenue	123 engagementcount	Bellevue	WA	98006	32,105	31	Auburn	WA	98002	25,585	8	Tacoma	WA	98413	25,490	19	Kirkland	WA	98033	19,705	20	Redmond	WA	98052	18,615	18	Seattle	WA	98105	10,795	8	Seattle	WA	98115	8,255	7
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		• Bellevue is the primary revenue hub, generating \$32,105 and 31																																										

		engagements - far ahead of any other city. • Auburn and Tacoma follow with \$25K revenue range and 8–19 engagements, showing consistent mid-level demand. • Seattle demand is fragmented: ◦ ZIP 98105 generates \$10,795 with 8 engagements, ◦ ZIP 98115 only \$8,255 from 7 engagements. → Different Seattle ZIPs have different customer value levels. • Kirkland & Redmond fall in the \$18–20K tier, stable but not high-value. Bellevue is the core market → prioritize top entertainers, marketing, and agent focus there. Seattle ZIP-level segmentation is required — customer value varies significantly within the city • Auburn emerged as the most significant geographic finding. With only 8 engagements generating \$25.6K, Auburn achieved a \$3,200 per-booking average, double the overall rate. While we lacked demographic data to explain this efficiency, the magnitude of the difference was too substantial to ignore. Bellevue's absolute revenue (\$32K) was higher, but Auburn's efficiency metrics made it the more compelling growth opportunity.
2	Repeat Customer Analysis Repeat customer = customers with more than one engagement We also want count of engagements, total revenue, first–last engagement span	-- 2. Repeat customer analysis <pre> SELECT c.CustomerID, c.CustFirstName ' ' c.CustLastName AS CustomerName, COUNT(e.EngagementNumber) AS EngagementCount, SUM(e.ContractPrice) AS TotalRevenue, MIN(e.StartDate) AS FirstEngagement, MAX(e.StartDate) AS LastEngagement FROM Customers c JOIN Engagements e ON c.CustomerID = e.CustomerID GROUP BY c.CustomerID, c.CustFirstName, c.CustLastName HAVING COUNT(e.EngagementNumber) > 1 ORDER BY EngagementCount DESC; </pre>

		<table><tr><th>customerid</th><th>customername</th><th>engagementcount</th><th>totalrevenue</th><th>firstengagement</th><th>lastengagement</th></tr><tr><td>10,010</td><td>Zachary Ehrlich</td><td>13</td><td>12,455</td><td>2017-09-19</td><td>2018-02-25</td></tr><tr><td>10,004</td><td>Dean McCrae</td><td>11</td><td>11,800</td><td>2017-09-12</td><td>2018-02-25</td></tr><tr><td>10,002</td><td>Deb Waldal</td><td>10</td><td>12,320</td><td>2017-09-30</td><td>2018-02-18</td></tr><tr><td>10,014</td><td>Mark Rosales</td><td>10</td><td>12,770</td><td>2017-09-11</td><td>2018-03-04</td></tr><tr><td>10,006</td><td>Matt Berg</td><td>9</td><td>13,170</td><td>2017-09-02</td><td>2018-02-24</td></tr><tr><td>10,009</td><td>Sarah Thompson</td><td>8</td><td>7,090</td><td>2017-09-30</td><td>2018-02-25</td></tr><tr><td>10,005</td><td>Elizabeth Hallmark</td><td>8</td><td>25,585</td><td>2017-09-16</td><td>2018-02-20</td></tr><tr><td>10,001</td><td>Doris Hartwig</td><td>8</td><td>10,795</td><td>2017-09-11</td><td>2018-02-24</td></tr><tr><td>10,003</td><td>Peter Brehm</td><td>7</td><td>7,250</td><td>2017-09-18</td><td>2018-02-27</td></tr><tr><td>10,012</td><td>Kerry Patterson</td><td>7</td><td>6,815</td><td>2017-10-01</td><td>2018-02-20</td></tr><tr><td>10,007</td><td>Liz Keyser</td><td>7</td><td>4,685</td><td>2017-09-12</td><td>2018-02-20</td></tr><tr><td>10,015</td><td>Carol Viescas</td><td>7</td><td>8,255</td><td>2017-10-08</td><td>2018-02-20</td></tr><tr><td>10,013</td><td>Estella Pundt</td><td>6</td><td>7,560</td><td>2017-10-15</td><td>2018-02-26</td></tr></table> - Many customers have 8–13 repeat bookings → very strong loyalty. - Repeat customers generate the highest lifetime revenue - Customer 10,005: 8 engagements → \$25,585 (highest in dataset). - Engagement dates span 2017 → 2018/2019, showing multi-year retention. - No customers appear to lapse early, most continue across multiple years. TuneWorks has excellent retention among core customers. High-value customers tend to book annually or multiple times per year, contributing predictable revenue.	customerid	customername	engagementcount	totalrevenue	firstengagement	lastengagement	10,010	Zachary Ehrlich	13	12,455	2017-09-19	2018-02-25	10,004	Dean McCrae	11	11,800	2017-09-12	2018-02-25	10,002	Deb Waldal	10	12,320	2017-09-30	2018-02-18	10,014	Mark Rosales	10	12,770	2017-09-11	2018-03-04	10,006	Matt Berg	9	13,170	2017-09-02	2018-02-24	10,009	Sarah Thompson	8	7,090	2017-09-30	2018-02-25	10,005	Elizabeth Hallmark	8	25,585	2017-09-16	2018-02-20	10,001	Doris Hartwig	8	10,795	2017-09-11	2018-02-24	10,003	Peter Brehm	7	7,250	2017-09-18	2018-02-27	10,012	Kerry Patterson	7	6,815	2017-10-01	2018-02-20	10,007	Liz Keyser	7	4,685	2017-09-12	2018-02-20	10,015	Carol Viescas	7	8,255	2017-10-08	2018-02-20	10,013	Estella Pundt	6	7,560	2017-10-15	2018-02-26
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3	Engagement patterns over months (using ztblMonths)	-- 3. Engagement patterns over months <pre>SELECT m.MonthYear, m.MonthNumber, COUNT(e.EngagementNumber) AS EngagementCount, SUM(e.ContractPrice) AS TotalRevenue FROM ztblMonths m LEFT JOIN Engagements e ON e.StartDate BETWEEN m.MonthStart AND m.MonthEnd GROUP BY m.MonthYear, m.MonthNumber ORDER BY m.MonthNumber;</pre>																																																																																				

A-Z monthyear	123 monthnumber	123 engagementcount	123 totalrevenue
January 2019	1	0	[NULL]
January 2018	1	28	43,880
January 2017	1	0	[NULL]
February 2018	2	20	21,685
February 2017	2	0	[NULL]
February 2019	2	0	[NULL]
March 2019	3	0	[NULL]
March 2017	3	0	[NULL]
March 2018	3	1	1,850
April 2019	4	0	[NULL]
April 2017	4	0	[NULL]
April 2018	4	0	[NULL]
May 2019	5	0	[NULL]
May 2017	5	0	[NULL]
May 2018	5	0	[NULL]
June 2017	6	0	[NULL]
June 2018	6	0	[NULL]
June 2019	6	0	[NULL]
July 2018	7	0	[NULL]
July 2017	7	0	[NULL]
July 2019	7	0	[NULL]
August 2019	8	0	[NULL]
August 2017	8	0	[NULL]
August 2018	8	0	[NULL]
September 2018	9	0	[NULL]
September 2019	9	0	[NULL]
September 2017	9	16	20,135
October 2019	10	0	[NULL]
October 2017	10	22	30,125
October 2018	10	0	[NULL]
November 2017	11	7	5,600
November 2018	11	0	[NULL]
November 2019	11	0	[NULL]
December 2018	12	0	[NULL]
December 2019	12	0	[NULL]
December 2017	12	17	17,275

- January is extremely high-volume, especially:
 - Jan 2018: 28 engagements, \$43,880 revenue.
- February also strong (20 engagements, \$21,685 in 2018).
- October is another peak month
 - Oct 2018: 22 engagements, \$30,125 revenue.
- Summer months (June to August) consistently show near-zero activity.
- Late-year activity (Nov–Dec) is inconsistent, mostly low except:
 - Dec 2017: 17 engagements, ~\$17K.

		Seasonality Interpretation • Q1 (Jan–Feb) = Strongest booking season. • Q4 (Oct) = strong secondary season. • Summer = low demand, customers do not book events mid-year. Business impact • Staffing and entertainer availability should be optimized for Jan–Feb + October. • The seasonality analysis was limited by data availability. January 2018 showed high activity (28 engagements, \$43K), and summer months appeared consistently weak, but with only six months of data we couldn't distinguish normal seasonal patterns from anomalous events. This represented a clear analytical limitation. TuneWorks would need at least 12 months of historical data before implementing staffing or pricing strategies based on seasonal trends.
4	Customer preferences → bookings correlation From Musical_Preferences table, get each customer's preference ranking Join with bookings to see whether preferences lead to actual bookings	-- 4. Preferences → bookings correlation <pre> SELECT mp.CustomerID, ms.StyleName, mp.PreferenceSeq, COUNT(e.EngagementNumber) AS BookingsForPreferredStyle FROM Musical_Preferences mp JOIN Musical_Styles ms ON mp.StyleID = ms.StyleID LEFT JOIN Engagements e ON e.CustomerID = mp.CustomerID LEFT JOIN Entertainer_Styles es ON e.EntertainerID = es.EntertainerID AND es.StyleID = mp.StyleID GROUP BY mp.CustomerID, ms.StyleName, mp.PreferenceSeq ORDER BY mp.CustomerID, mp.PreferenceSeq; </pre>

123 customerid	AZ stylename	123 preferenceseq	123 bookingsforpreferredstyle
10,001	Top 40 Hits	1	8
10,001	Contemporary	2	8
10,002	60's Music	1	10
10,002	Classic Rock & Roll	2	10
10,003	Rhythm and Blues	1	7
10,003	Motown	2	7
10,004	Jazz	1	11
10,004	Standards	2	11
10,005	Chamber Music	1	8
10,005	Classical	2	8
10,006	Variety	1	9
10,006	Folk	2	9
10,007	Classic Rock & Roll	1	7
10,007	70's Music	2	7
10,007	Rhythm and Blues	3	7
10,008	Contemporary	1	0
10,008	Standards	2	0
10,009	Country Rock	1	8
10,009	Country	2	8
10,009	Modern Rock	3	8
10,010	Rhythm and Blues	1	13
10,010	Jazz	2	13
10,010	Salsa	3	13
10,011	Standards	1	0
10,011	40's Ballroom Music	2	0
10,011	Classical	3	0
10,012	Show Tunes	1	7
10,012	Contemporary	2	7
10,013	Jazz	1	6
10,013	Salsa	2	6
10,014	Top 40 Hits	1	10
10,014	Modern Rock	2	10
10,014	80's Music	3	10
10,015	Standards	1	7
10,015	Show Tunes	2	7
10,015	40's Ballroom Music	3	7

- Customers *do* frequently book their top-preferred style:
 - Customer *10,010*: Prefers Rhythm and Blues (Seq 1) → books 13 times.
 - Customer *10,001*: Prefers Top 40 Hits → books 8 times.
 - Customer *10,014*: Prefers Top 40 Hits → books 10 times.
- Lower preference ranks (seq 2–3) still show booking activity, but at a much lower frequency.
- Some styles are preferred but rarely booked, indicating:
 - Availability mismatch, or
 - Customer interest but not enough entertainer supply.
- TuneWorks should match entertainer inventory to customer preferences.

		- Top 40 Hits, R&B, Jazz, Contemporary show actual booking alignment → reliable demand drivers. - Styles with preference but zero bookings represent latent demand or inventory gaps.																																										
5	High-value customer segments Definition: top 25% by total revenue use NTILE(4))	-- 5. High-value customer segments <pre> WITH cust_rev AS (SELECT c.CustomerID, SUM(e.ContractPrice) AS TotalRevenue FROM Customers c JOIN Engagements e ON c.CustomerID = e.CustomerID GROUP BY c.CustomerID) SELECT CustomerID, TotalRevenue, NTILE(4) OVER (ORDER BY TotalRevenue DESC) AS RevenueQuartile FROM cust_rev ORDER BY TotalRevenue DESC; </pre> <table border="1"> <thead> <tr> <th>123 customerid</th><th>123 totalrevenue</th><th>123 revenuequartile</th></tr> </thead> <tbody> <tr><td>10,005</td><td>25,585</td><td>1</td></tr> <tr><td>10,006</td><td>13,170</td><td>1</td></tr> <tr><td>10,014</td><td>12,770</td><td>1</td></tr> <tr><td>10,010</td><td>12,455</td><td>1</td></tr> <tr><td>10,002</td><td>12,320</td><td>2</td></tr> <tr><td>10,004</td><td>11,800</td><td>2</td></tr> <tr><td>10,001</td><td>10,795</td><td>2</td></tr> <tr><td>10,015</td><td>8,255</td><td>3</td></tr> <tr><td>10,013</td><td>7,560</td><td>3</td></tr> <tr><td>10,003</td><td>7,250</td><td>3</td></tr> <tr><td>10,009</td><td>7,090</td><td>4</td></tr> <tr><td>10,012</td><td>6,815</td><td>4</td></tr> <tr><td>10,007</td><td>4,685</td><td>4</td></tr> </tbody> </table> - Quartile 1 customers (top 25%): - Generate \$12,455 – \$25,585 each. - Book 10+ engagements consistently. - Are highly loyal, multi-year clients. - Quartile 2 customers still strong (\$10–12K), mid-loyalty (7–11 engagements).	123 customerid	123 totalrevenue	123 revenuequartile	10,005	25,585	1	10,006	13,170	1	10,014	12,770	1	10,010	12,455	1	10,002	12,320	2	10,004	11,800	2	10,001	10,795	2	10,015	8,255	3	10,013	7,560	3	10,003	7,250	3	10,009	7,090	4	10,012	6,815	4	10,007	4,685	4
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		• Quartile 3: decreasing engagement, sometimes low frequency. • Quartile 4: typically \$4–7K revenue contributors, lowest engagement count. • Q1 customers are the financial backbone → must be prioritized. • Q2 customers show potential → can be converted into Q1 with targeted retention offers. • Q4 customers may not justify heavy personalization — cost-conscious or infrequent bookers.
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Phase 4: Synthesis & Recommendations

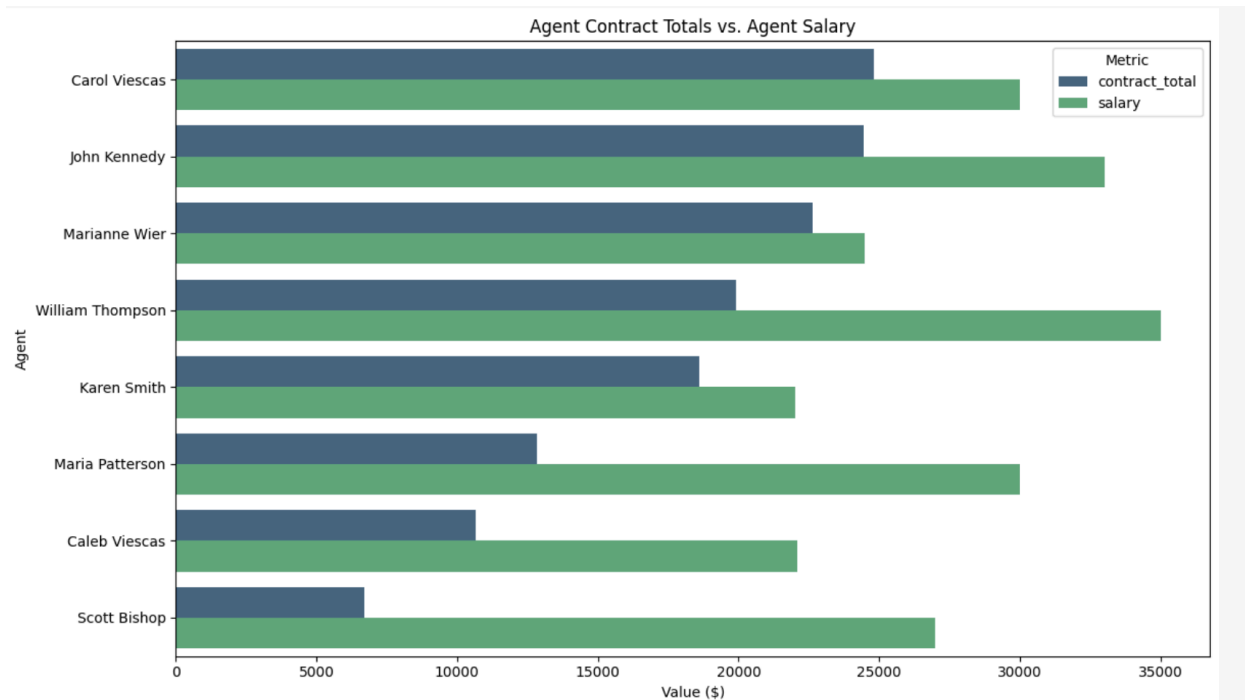
Introduction

The overall objective of this synthesis and recommendation section is to identify insights and their corresponding recommendations for the major operational pillars of TuneWorks. In general, KPIs across different categories varied highly. The challenge: performance is highly concentrated in a few key agents, entertainers, and markets, creating a capacity ceiling that limits long-term growth. Our proposed actions seek to build upon TuneWorks’ current strengths while strategically developing additional sources of revenue in underutilized areas. By following these recommendations, TuneWorks will be able to improve operations and revenue, identify opportunities, and especially plan for long-term sustainable growth.

Agent Analysis

Insights

Analysis of the six months of engagement data shows that Carol, John, and Marianne are the top-performing agents based on total contract value. In contrast, Scott is the most notable underperformer relative to his compensation: despite low performance, he has the fifth-highest base salary and commission rate. Additional performance issues include Daffy, who has no output; Scott and Caleb, who both generate low engagement volume; and Maria, who secures many contracts but does so at a significantly lower average price (\$855 compared to over \$1,000 for almost all other agents). High-value clients are currently served by the strongest agents. While this alignment should continue, developing additional agent capacity will enable TuneWorks to handle increased volume and reduce dependence on a few top performers.



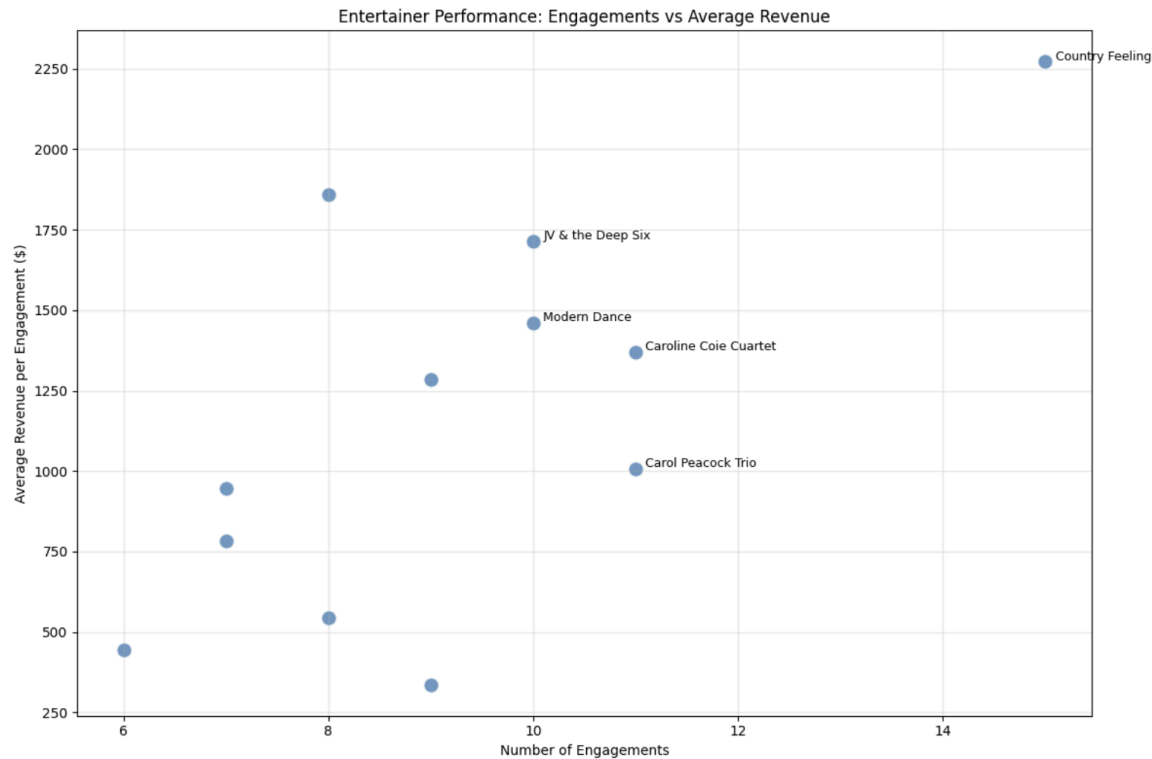
Recommendations

Compensation structures should be realigned so that commission rates, and potentially base salaries, correspond more directly with demonstrated performance. Agents exceeding \$25K in quarterly bookings receive 15% commission on incremental revenue, driving competitive performance. These adjustments should be made once a full 12 months of data is available to ensure accuracy. Targeted training should also be implemented: Daffy would benefit from onboarding support, Scott and Caleb need guidance in lead generation and sales pipeline development, and Maria should receive coaching to strengthen her negotiation and pricing strategies. Because Carol, John, and Marianne consistently generate strong results, they should continue receiving priority access to the highest-value leads.

Entertainer & Style Analysis

Insights

Country Feeling stands out as the strongest entertainer, achieving both the highest number of engagements (15) and the highest average revenue per engagement. Several other acts, including Susan McLain, Jim Glynn, Julia Schnebly, and Jazz Persuasion, produce minimal revenue. From a stylistic standpoint, 60's Music and Country are the dominant and most commercially successful genres, while many other genres, such as Rap, Modern Rock, Ballroom, Elvis, and various decade-based genres, generate no revenue at all.



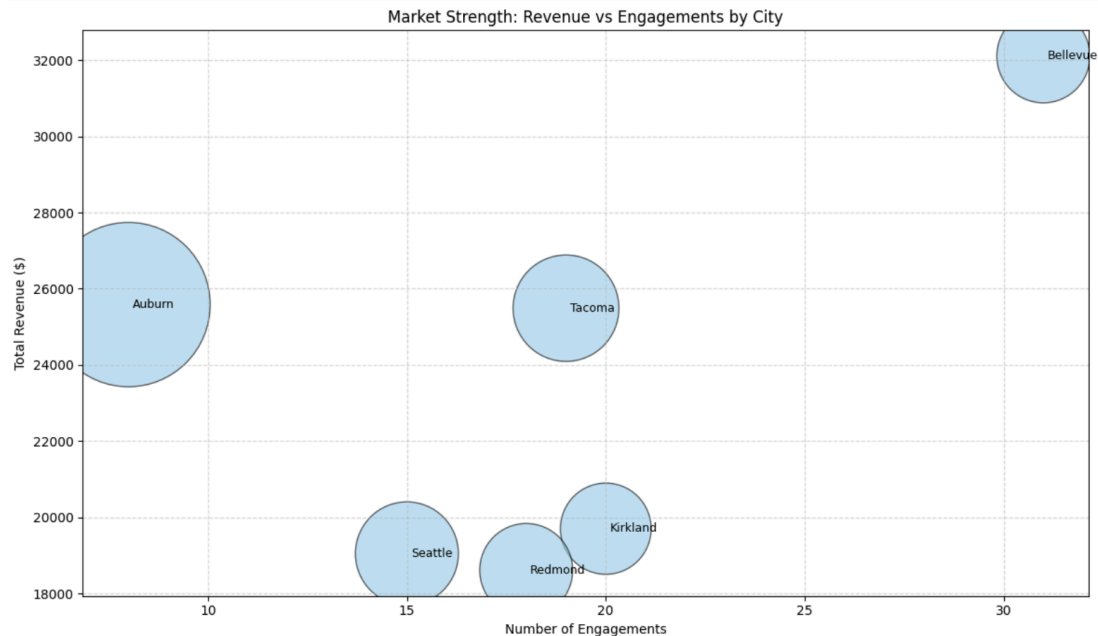
Recommendations

The agency should continue prioritizing Country Feeling (\$34K), JV & the Deep Six (\$17K), and Caroline Coie Cuartet (\$15K), the top revenue-generating entertainers. To reduce dependence on individual performers and build additional capacity, TuneWorks should pursue two parallel strategies: (1) actively recruit 2-3 new acts within proven high-demand genres (60's Music and Country, which generate 43% of total revenue), and (2) invest marketing resources in developing mid-tier performers like Coldwater Cattle Company, Modern Dance, and Saturday Revue, who show solid performance but could benefit from increased promotion. Low-performing acts generating minimal revenue should be evaluated for repositioning or removal, with resources redirected to support these growth initiatives.

Customer & Demand Trends

Insights

Bellevue emerges as the strongest market in both revenue and engagement count. Auburn also shows exceptional promise, generating \$25.6k from just eight engagements, indicating a highly efficient revenue-to-engagement ratio. Tacoma is the third-strongest city and remains an important market. Customer behavior shows strong loyalty, with many clients booking repeatedly and typically favoring the same genres across engagements. Due to limited data, it is difficult to draw conclusions about seasonality.



Recommendations

Efforts should be focused on maintaining and expanding the Bellevue market while investing further in Auburn, where increased engagement volume could yield substantial returns. Tacoma should also be prioritized for growth marketing. Advertising should be tailored to the specific genres preferred by different demographic groups, with older audiences, for example, more likely to respond to promotions for 60's Music. Genre-relevant conventions and events should be considered for targeted outreach. Additional months of data collection are needed before implementing seasonality-based staffing or forecasting decisions.

Summary

Overall, TuneWorks should continue investing in top performers while developing additional revenue sources in underutilized areas. The most important recommendations for TuneWorks are to realign agent compensation and training to better match actual performance, prioritize top-performing entertainers and high-demand genres, and focus marketing resources on the strongest geographic markets. Compensation and commission structures should be revised once a full year of data is available, while underperforming agents receive targeted skill development. The agency should deepen its relationship with Country Feeling and expand offerings within 60's Music and Country. Marketing efforts should center on Bellevue, Auburn, and Tacoma, with targeted outreach based on genre and demographic fit. Finally, improved data collection, especially around revenue streams, operational expenses, customer demographics, and agent KPIs, will enable more accurate performance evaluation and more strategic decision-making going forward. By balancing investment in current strengths with development of new capacity, TuneWorks can build a more resilient, scalable business positioned for sustainable long-term growth.